

Linear Algebra Homework

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1 HW Due Mar 10

Problem 1.1. You will get full credit for this problem as long as you submit this HW, no matter how you answer these questions.

1. What is your expected letter grade for this class?
2. How many hours per week do you plan to commit for this class?
3. Name a friend in this class you could discuss math with.
4. Name a friend that has already aced this class so you can ask questions from.
5. Name your favorite AI model as your personal assistant in all matters. (If you can afford it, use a paid version of whatever AI model. It is worth the investment in my opinion.)

Problem 1.2. We interpret $\begin{bmatrix} x \\ y \end{bmatrix}$ as an arrow vector.

1. Move the arrows $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ -1 \end{bmatrix}$ around to form a triangle.
2. You can move the arrows $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$ around, but they will never form a triangle. Why? (Hint: Check their lengths.)
3. Find some coefficients a, b, c such that

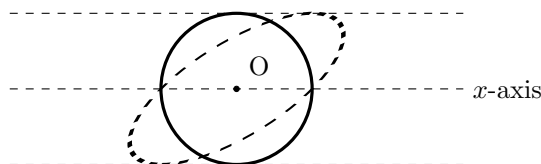
$$a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

4. Find a triangle whose three sides are parallel to the vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$. Maybe use the last subproblem as a inspiration? (Remark: the answer here is not unique, but all possible answers are similar triangles. As an extra challenge, can you see why? There is an algebraic reason and a geometric reason.)
5. You can move the arrows $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ around, but they will never form a triangle. Why?

Problem 1.3. Find the matrices of the following linear planar transformations.

1. Let α be the line obtained by rotating x -axis around the origin counterclockwise by $\frac{\pi}{8}$. Find the matrix for the reflection about the line α .
2. Find the matrix for the rotation around the origin, that sends $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to $\begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$.
3. Find the matrix for the linear map that fixes all the points on x -axis, but stretches the plane along the y -axis by a factor of 3.
4. Find the matrix for the linear map that fixes all the points on the line $x + y = 0$, but stretches the plane along the direction of $x = y$ by a factor of 2.
5. Find the matrix for the linear map that sends $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ to $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ to $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$. (Hint: This map is linear. So where should it send $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$?)

6. Find a matrix for the linear map that sends the circle below to the dashed ellipse. Here the point $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is fixed, and the point $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is sent to the point $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.



Problem 1.4. Are these matrix-vector multiplication defined? If it is defined, calculate the result of the matrix-vector multiplication.

1. $\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

2. $\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

3. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

4. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

5. $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

6. $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

7. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$. (Can you see a pattern?)

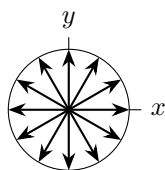
Problem 1.5 (Matrix-Vector Multiplication). 1. Let us define a ***magic matrix*** to be a 3×3 matrix whose entries are $1, \dots, 9$ in some order, such that in each row, each column, and each of the two diagonals, the three entries add up to the same number. For example, a typical magic matrix is $\begin{bmatrix} 2 & 9 & 4 \\ 7 & 5 & 3 \\ 6 & 1 & 8 \end{bmatrix}$. For an arbitrary magic matrix M , try to find all possible values of $M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

2. Let us define a ***Sudoku matrix*** to be a 9×9 matrix such that each row is made of $1, \dots, 9$, and each column is made of $1, \dots, 9$, and each of the nine 3×3 “submatrix” is also made of $1, \dots, 9$. For

example, a typical Sudoku matrix is

5	3	4	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6	1	5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

For an arbitrary Sudoku matrix M , try to find all possible values of $M \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$.



Problem 1.6. Look at this clock. There are twelve arrow vectors here. Assume that they are all unit vectors on the xy -plane.

1. Find the sum of all twelve vectors.
2. Find the sum of all vectors except the 2 o'clock vector.
3. Fix the end points of all vectors, but move the starting points of all vectors to the 6 o'clock location, i.e., the 12 o'clock vector is now $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$, and the 3 o'clock vector is now $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and so on. Now find the sum of all twelve vectors.

Problem 1.7. Write the following vector $\mathbf{b}$ as the result of a matrix multiplying a vector.

1. $\mathbf{b} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 5 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

2. $\mathbf{b} = 5 \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} + 4 \begin{bmatrix} 5 \\ 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}$.

3. $\mathbf{b} = \begin{bmatrix} 2b + a + c \\ c - b \\ a + b + c \\ a + b \end{bmatrix}$, where a, b, c are constants.

4. $\mathbf{b} = f \left(\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} \right)$, where $f : \mathbb{R}^5 \rightarrow \mathbb{R}^5$ is a linear map such that $f(\mathbf{e}_k) = k\mathbf{e}_{6-k}, k = 1, \dots, 5$.

5. ☛ Suppose we know the following rule of the weathers. If it rains one day, then the next day has 0.8 chance of raining. If it does not rain one day, then the next day has 0.3 chance of raining. Suppose today has 0.5 chance of raining, let $\mathbf{b} = \begin{bmatrix} p \\ 1 - p \end{bmatrix}$ where p is the chance of raining tomorrow.

2 HW Due Mar17

Problem 2.1. Find the RREF of the following matrices.

1. $\begin{bmatrix} -1 & 1 \\ -1 & 0 \\ 20 & 300 \\ 1234 & 5678 \end{bmatrix}$. This first one has no credit, I will do it for you to show you the acceptable format.

Method one, Gaussian elimination. If you do the standard Gaussian elimination, then you can simply write

$$\begin{bmatrix} -1 & 1 \\ -1 & 0 \\ 20 & 300 \\ 1234 & 5678 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 \\ 0 & -1 \\ 0 & 320 \\ 0 & 6912 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 \\ 0 & -1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

In short, when you use a pivot to kill all entries below, then you can combine that into a single step, and no need to elaborate on the specifics of the row operation because you are following the standard Gaussian elimination algorithm. We all know the moves you are doing.

Method two, your own smarter elimination. Then you need to indicate your steps. For example

$$\begin{bmatrix} -1 & 1 \\ -1 & 0 \\ 20 & 300 \\ 1234 & 5678 \end{bmatrix} \xrightarrow{\begin{matrix} r_1 \rightarrow r_1 - r_2 \\ r_3 \rightarrow r_3 + 20r_2 \\ r_4 \rightarrow r_4 - 1234r_2 \end{matrix}} \begin{bmatrix} 0 & 1 \\ -1 & 0 \\ 0 & 300 \\ 0 & 5678 \end{bmatrix} \xrightarrow{\begin{matrix} r_3 \rightarrow r_3 - 300r_1 \\ r_4 \rightarrow r_4 - 5678r_1 \end{matrix}} \begin{bmatrix} 0 & 1 \\ -1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \xrightarrow{r_1 \rightarrow -r_1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

In this case, you must explain your steps. If you are not doing Gaussian elimination, then it is NOT clear to me what you are doing, so you need to be more specific. Also note that you can ONLY combine moves into one arrow if they commute with each other (order of moves must have no effect within this step)! If you combine moves that fail to commute, we shall treat that as wrong. For example, the following is a WRONG step

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \xrightarrow{\begin{matrix} r_2 \rightarrow r_2 - r_1 \\ r_1 \rightarrow r_1 - r_2 \end{matrix}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

This is wrong because doing $r_2 \rightarrow r_2 - r_1$ first and then $r_1 \rightarrow r_1 - r_2$ is DIFFERENT from doing $r_1 \rightarrow r_1 - r_2$ first and then $r_2 \rightarrow r_2 - r_1$. I can never be sure which order you mean. (More importantly, I cannot instantly compute combinations of non-commuting moves in my head, so I would have to write out the intermediate steps myself to see if you are right. I can combine commuting moves in my head easily though.)

Finally, method three. You can spot the RREF from column relations alone. To do this, you need to write “The first column is non-zero, so it is going to be a pivot column in the RREF. The second column is not parallel to the first column, so it is not a linear combination of the previous column.

So the second column is also a pivot column in RREF. Hence the RREF is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$.” You need to

explicitly explain why each column is not a linear combination of previous columns, or it is which linear combination of previous columns.

2. $\begin{bmatrix} 1 & 3 & 2 & 1 \\ 2 & -3 & 0 & -2 \end{bmatrix}$.

$$3. \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

$$4. \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}.$$

$$5. \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 11 \end{bmatrix}.$$

$$6. \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 7 \\ 7 & 8 & 9 \end{bmatrix}.$$

$$7. \begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

Problem 2.2. Consider the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$. Only use row scaling and row shearing, can you turn it into $\begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$? Show me how.

Problem 2.3 (Gaussian Eliminations). Solve these by Gaussian elimination, and write out the solution set explicitly. (So don't just stop at RREF for this problem. Write out the set, something like $\begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix} + x \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} +$

$z \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} : x, z \text{ free}$ in our example in class.)

$$1. \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}.$$

$$2. \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}.$$

$$3. \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 8 \end{bmatrix}.$$

Problem 2.4 (Situations of the solution set). Find a constant b that satisfies the condition.

$$1. \begin{bmatrix} 1 & 2 \\ 3 & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix} \text{ has no solution.}$$

$$2. \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ b \end{bmatrix} \text{ has no solution.}$$

3. $\begin{bmatrix} 2 & 5 & 1 \\ 4 & b & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$ has no solution.
4. $\begin{bmatrix} b & 3 \\ 3 & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ -6 \end{bmatrix}$ has infinitely many solutions.
5. $\begin{bmatrix} 2 & b \\ 4 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 16 \\ c \end{bmatrix}$ has infinitely many solutions. (For this problem, find constants b, c that satisfy the condition.)
6. $\begin{bmatrix} 1 & b & 0 \\ 1 & -2 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ has a non-zero solution.
7. $\begin{bmatrix} b & 2 & 3 \\ b & b & 4 \\ b & b & b \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ has a non-zero solution. (For this problem, find three possible values for b that satisfies the condition.)

Problem 2.5 (Solving a hardcore linear system). ♣ For each real value of p , find all solutions to the following system:

$$\begin{bmatrix} p & 1 & 1 \\ 1 & p & 1 \\ 1 & 1 & p \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ p \\ p^2 \end{bmatrix}.$$

(Hint: first do $r_1 \rightarrow r_1 + r_2 + r_3$, and then do $r_1 \rightarrow \frac{1}{p+1}r_1$. However, what if $p = -1$? I guess you have to break it down into different cases for this discussion.)

Problem 2.6 (Rank-one matrices). If a matrix has all rows parallel, must it also have all columns parallel? (As a convention, we treat $\mathbf{0}$ as parallel to everything.) Give a counter-example or prove why.

(Hint: This problem is called rank-one matrices. Guess why....)

3 HW Due Mar 24

Problem 3.1 (Arithmetic sequences in Matrices). For a given matrix, suppose the columns are all arithmetic

sequences. (For example, something like $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 3 \\ 5 & 4 & 2 \\ 7 & 5 & 1 \end{bmatrix}$.)

1. Assume that the first row of such a matrix is $\mathbf{a}^T$ and the second row is $\mathbf{b}^T$. Can you deduce all other rows from $\mathbf{a}^T$ and $\mathbf{b}^T$?
2. Show that such a matrix must have rank at most 2.

Problem 3.1 (Dimensions in matrix multiplications). Say A, B, C are $3 \times 5, 5 \times 3, 3 \times 1$ matrices respectively, then which of $BA, AB, ABAB, BAC, BABC$ are well-defined?

Problem 3.2. Calculate the following matrix multiplications. Or maybe it is undefined!

1. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$.

2. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

3. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

4. $\begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

5. $\begin{bmatrix} 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

6. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

7. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} [2]$.

8. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} [1 \quad 2 \quad 3 \quad 4]$.

9. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$.

10. $\begin{bmatrix} 1 & 2 & 1 \\ & 1 & 1 \\ & & 1 \end{bmatrix} \begin{bmatrix} x^2 \\ x \\ 1 \end{bmatrix}$.

$$11. \begin{bmatrix} 1 & 2 & 1 \\ & 1 & 1 \\ & & 1 \end{bmatrix} \begin{bmatrix} (x+1)^2 \\ x+1 \\ 1 \end{bmatrix}.$$

$$12. \begin{bmatrix} 1 & 2 & 1 \\ & 1 & 1 \\ & & 1 \end{bmatrix} \begin{bmatrix} (x+2)^2 \\ x+2 \\ 1 \end{bmatrix}. \text{ (Do you see a pattern here?)}$$

Problem 3.3. Calculate the following matrix multiplications.

$$1. \begin{bmatrix} 1 & 1 & 1 \\ & 1 & 1 \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}. \text{ (Can you generalize this to an } n \times n \text{ pattern?)}$$

$$2. \begin{bmatrix} & & 1 \\ & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 \\ 1 \end{bmatrix}. \text{ (Can you generalize this to an } n \times n \text{ pattern?)}$$

$$3. \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}^2. \text{ (Can you generalize this to an } n \times n \text{ pattern?)}$$

$$4. \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}^2. \text{ (Can you generalize this to an } n \times n \text{ pattern? All diagonal entries are 1 and off-diagonal entries are } -1\text{.)}$$

$$5. \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}^2. \text{ (Can you generalize this to an } n \times n \text{ pattern? All diagonal entries are 1, and adjacent entries alternate between } \pm 1\text{.)}$$

Problem 3.4. Calculate the following matrix multiplications.

1. Suppose $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are mutually orthogonal unit vectors in $\mathbb{R}^5$. Compute

$$\begin{bmatrix} \mathbf{u}^T \\ \mathbf{v}^T \\ \mathbf{w}^T \end{bmatrix} \begin{bmatrix} \mathbf{u} & \mathbf{v} & \mathbf{w} \end{bmatrix}.$$

$$2. \text{ Compute } \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}^2. \text{ (This is actually the square of a reflection.)}$$

$$3. \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix} \begin{bmatrix} \cos \phi & \sin \phi \\ \sin \phi & -\cos \phi \end{bmatrix}. \text{ This is a product of two reflections, and the result is a rotation. What is the angle of this rotation? (Can you also find a geometric proof?)}$$

Problem 3.5 (Powers of a matrix). Calculate the following matrix powers. (Maybe keep track of some patterns to help you do the large powers.)

$$1. \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^2.$$

$$2. \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^3.$$

$$3. \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^{2020}.$$

$$4. \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^{-1}.$$

$$5. \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}^2.$$

$$6. \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}^3.$$

$$7. \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}^{2020}.$$

$$8. \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}^2.$$

$$9. \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}^3.$$

$$10. \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}^{2020}.$$

$$11. \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}^{-1}.$$

Problem 3.6 (Polynomials of a matrix). Suppose for a matrix A we have $A^2 = A$.

1. Simplify the polynomials $A^3 + 2A^2 - A - I$ and $A^2 + 3A + 4I$ into the format $sA + tI$ for some constants $s, t \in \mathbb{R}$. (Read only: note that you can use this strategy to simplify all possible polynomials of A into the format $sA + tI$.)
2. 🐼 Show that $I + 2A$ is invertible by finding its inverse. (Hint: Suppose that the inverse is $xI + yA$, then solve x, y from $(I + 2A)(xI + yA) = I$. Use $A^2 = A$ to simplify.)

4 HW Due Mar 31

Problem 4.1 (Commutativity trouble). Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$.

1. Calculate AB and BA . Are they the same?
2. Calculate $(A + B)^2$, $A^2 + 2AB + B^2$. Are they the same?
3. Calculate $(AB)^2$, A^2B^2 . Are they the same?

Problem 4.2 (Polynomials of a matrix). (Ignore this whole problem!!!! Repeated from last HW....)

Suppose for a matrix A we have $A^2 = A$.

1. Simplify the polynomials $A^3 + 2A^2 - A - I$ and $A^2 + 3A + 4I$ into the format $sA + tI$ for some constants $s, t \in \mathbb{R}$. (Read only: note that you can use this strategy to simplify all possible polynomials of A into the format $sA + tI$.)
2. 🐛 Show that $I + 2A$ is invertible by finding its inverse. (Hint: The inverse is also a polynomial of A . Use the last subproblem to simplify your assumption!)

Problem 4.3 (Matrix multiplications and row/column operations). Calculate the following matrix multiplications

1. $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$.
2. $\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 1 \\ 1 & 4 & 0 \end{bmatrix}$.
3. $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 \\ 1 & 3 & 3 & 1 & 0 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$.
4. $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}^k$ for an integer k . Find the formula depending on k .

Problem 4.4 (Elementary shearings minus identity). In $\mathbb{R}^4$, let $X_{ij} = \mathbf{e}_i \mathbf{e}_j^T$ be the corresponding 4 by 4 matrix. (Note that the elementary shearing matrix is $E_{ij}^k = I + kX_{ij}$.)

1. Write out X_{13}, X_{32}, X_{12} .
2. Calculate $X_{13}X_{32}$ and $X_{32}X_{13}$. (See if you can do this via the fact that $X_{ij} = \mathbf{e}_i \mathbf{e}_j^T$, and use associativity. Dot products between these standard basis vectors are super easy.)
3. Verify that $X_{ij}^2 = O$ when $i \neq j$. (Not part of the problem, but you may realize that this implies that $(E_{ij}^1)^k = (I + X_{ij})^k = I + kX_{ij} = E_{ij}^k$.)
4. For any two square matrices A, B , show that $AB = BA$ if and only if $(A - I)(B - I) = (B - I)(A - I)$. (Hence to study the commutativity behavior between elementary shearings, we only need to study the commutativity of these X_{ij} .)

Problem 4.5 (Shifting operators). Let $J = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

1. Going from A to JA , would J be shifting things up? down? left? right?
2. Going from A to AJ , would J be shifting things up? down? left? right?
3. Given the Pascal's symmetric matrix $P = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix}$, calculate $PJ + J^T P$. Can you explain the "coincidence" that most entries are identical? (Hint: understand the meaning behind $PJ + J^T$, and compare this meaning to the way P is defined.)
4. Calculate J, J^2, J^3, J^4 . (Also note that their appearances is exactly how they would shift the identity matrix.)
5. Show that $I - J$ has inverse $I + J + J^2 + J^3$.
6. Calculate $(J + I)^2, (J + I)^3, (J + I)^4$. (Not part of the problem, but can you see a pattern for the entries? Can you prove the pattern for $(J + I)^k$?)
7. ♣ Describe the set of all matrices that commutes with J .

Problem 4.6 (Gaussian elimination to find inverse). Find inverse matrices. You may start with $[A \ I_n]$ and work your way to $[I_n \ A^{-1}]$, or you may do whatever that works.

1. $\begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$.

2. $\begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$. (Compare this with above. Merely a single-entry difference, yet the inverses look drastically different. However, is the difference rank one?)

3. $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. (Horses move like this in chess. This pattern is also useful in five-in-a-row, i.e., Wü Zǐ Qí.)

4. $\begin{bmatrix} & & & 1 \\ & & \ddots & \\ & & & \\ 1 & & & \end{bmatrix}$.

5. $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 4 \end{bmatrix}$.

$$6. \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}.$$

$$7. \begin{bmatrix} 1 & -a & 0 & 0 \\ 0 & 1 & -b & 0 \\ 0 & 0 & 1 & -c \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Problem 4.7 (Constraints through matrix multiplications). In each cases, find a matrix B that satisfies the given condition.

1. $AB = 4A$ for all 3 by 3 matrices A .
2. $BA = 4A$ for all 3 by 3 matrices A .
3. Every row of BA is the first row of A for all 3 by 3 matrices A .
4. Every entry of AB is the average of the entries in the corresponding row of A for all 3 by 3 matrices A .
5. $B^2 \neq O$, $B^3 = O$. (Just find any example of such a matrix is enough.)
6. $B \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} B$. Find all possible B for this problem.

5 HW Due Apr 7

Problem 5.1 (Weird “triangular” matrices). An $n \times n$ matrix is called a **northwest** matrix if all entries below the anti-diagonal from $(1, n)$ entry to $(n, 1)$ entry are zero. Similarly, we can define a **southeast** matrix. Now, if A is northwest, what about A^T, A^{-1}, A^2 ? What if we multiply a northwest matrix with a

southeast matrix? (Hint: Let $F = \begin{bmatrix} & & 1 \\ & \ddots & \\ 1 & & \end{bmatrix}$, then what is F times a northwest matrix? Try to convert everything into upper/lower triangular matrices to see better.)

Problem 5.2. Calculate the following matrix multiplications. (Maybe try block matrix multiplications to get some feelings.)

$$1. \begin{bmatrix} 1 & 1 & & \\ & 1 & & \\ & & 1 & \\ & & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & & \\ & 1 & & \\ & & 1 & \\ & & 1 & 1 \end{bmatrix}.$$

$$2. \begin{bmatrix} 1 & 1 & -1 & \\ & 1 & & -1 \\ -1 & & 1 & -1 \\ & -1 & & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 2 & 3 \\ 2 & 3 & 5 & 7 \\ 1 & 1 & 2 & 3 \end{bmatrix}.$$

$$3. \begin{bmatrix} 1 & & & \\ & 1 & & \\ 1 & & -1 & \\ 1 & 1 & & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 \\ 6 & 8 & 10 & 12 \end{bmatrix}.$$

Problem 5.3. Calculate the following block matrix multiplications if possible, or decide that they are undefined.

$$1. \begin{bmatrix} A & B \\ B & A \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}. \text{ Assume } Av = e_1, Aw = e_2, Bv = e_3, Bw = e_4, \text{ and } A, B \text{ have four rows each.}$$

$$2. \begin{bmatrix} v \\ w \end{bmatrix} \begin{bmatrix} v & w \end{bmatrix}. \text{ All vectors are in } \mathbb{R}^2.$$

$$3. \begin{bmatrix} v & w \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}. \text{ All vectors are in } \mathbb{R}^2.$$

$$4. \begin{bmatrix} A & \\ A & A \end{bmatrix} \begin{bmatrix} A & A \\ A & \end{bmatrix} \text{ where } A \text{ is } 2 \times 2.$$

$$5. \begin{bmatrix} A & \\ A & A \end{bmatrix} \begin{bmatrix} A & A \\ A & \end{bmatrix} \text{ where } A \text{ is } 2 \times 3.$$

Problem 5.4 (Block Operations). Calculate the inverse of the following matrices. (Try not to use the ugly formula. Try do block operations from scratch for practice.)

$$1. \begin{bmatrix} I_n & O \\ A & I_m \end{bmatrix}. \text{ (This is lower triangular.)}$$

$$2. \begin{bmatrix} O & I_m \\ I_n & A \end{bmatrix}. \text{ (This is block-southeast.)}$$

$$3. \begin{bmatrix} O & A \\ B & O \end{bmatrix} \text{ where } A, B \text{ are invertible.}$$

4. $\begin{bmatrix} A & C \\ O & B \end{bmatrix}$ where A, B are invertible. (This is block upper triangular.)

Problem 5.5. Suppose the RREF of A is R , and the rows of A are linearly independent. Find the RREF of the following matrices.

1. $\begin{bmatrix} A & 2A \end{bmatrix}$.
2. $\begin{bmatrix} A \\ 2A \end{bmatrix}$.
3. $\begin{bmatrix} A & A \\ A & A \end{bmatrix}$.
4. $\begin{bmatrix} A & A \\ O & A \end{bmatrix}$, where the rank of A is equal to the number of rows of A .

Problem 5.6 (We can study affine maps using matrices as well). A map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is called an **affine map** if you can find an $n \times n$ matrix A and a vector $\mathbf{b} \in \mathbb{R}^n$ such that $f(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$. (It does not fix the origin, so it is not linear, merely affine. For example, $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = ax + b$ is an affine map.)

1. Show that $\begin{bmatrix} A & \mathbf{b} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} = \begin{bmatrix} f(\mathbf{x}) \\ 1 \end{bmatrix}$. Let us define M_f as $\begin{bmatrix} A & \mathbf{b} \\ \mathbf{0}^T & 1 \end{bmatrix}$, and call this the matrix for the affine map.
2. Show that $M_f M_g = M_{f \circ g}$.
3. Show that f is invertible iff A is invertible, and $M_f^{-1} = M_{f^{-1}}$. Also find the block inverse M_f^{-1} .

Problem 5.7 (Hidden Block Diagonal Matrix). Define $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \triangle \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11} & 0 & a_{12} & 0 \\ 0 & b_{11} & 0 & b_{12} \\ a_{21} & 0 & a_{22} & 0 \\ 0 & b_{21} & 0 & b_{22} \end{bmatrix}$

1. Find a matrix X such that for all A, B , $X(A \triangle B)X^{-1} = \begin{bmatrix} A & O \\ O & B \end{bmatrix}$.
2. Show that $(A_1 \triangle A_2)(B_1 \triangle B_2) = (A_1 B_1) \triangle (A_2 B_2)$. (The first sub-problem might help.)
3. Show that when A, B are invertible, $(A \triangle B)^{-1} = A^{-1} \triangle B^{-1}$. (The first sub-problem might help.)

6 HW Due Apr 14

Problem 6.1. The **trace** of an $n \times n$ matrix A is the sum of all of its diagonal entries. We denote it as $\text{tr}(A)$. If you like, you can write $\text{tr}(A) = \sum_{i=1}^n \mathbf{e}_i^T A \mathbf{e}_i$.

1. Show that $\text{tr}(xA + yB) = x \text{tr}(A) + y \text{tr}(B)$.
2. Show that $\text{tr}(I_n) = n$.
3. Show that $\text{tr}(A) = \text{tr}(A^T)$.
4. For a unit vector $\mathbf{u}$, show that $\text{tr}(\mathbf{u}\mathbf{u}^T) = 1$ and $\text{tr}(I_n - \mathbf{u}\mathbf{u}^T) = n - 1$. (For projections, trace simply tells you the dimension of the projection range.)
5. Show that, if $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$, $B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$, then $\text{tr}(A^T B) = \text{tr}(B A^T) = \sum a_i b_i$. (This is a “dot product” for matrices.)
6. Show that $\text{tr}(\mathbf{v}\mathbf{w}^T) = \text{tr}(\mathbf{w}^T \mathbf{v})$.
7. 🐛 Can you prove that $\text{tr}(AB) = \text{tr}(BA)$ in general? Here A is $m \times n$ and B is $n \times m$. (Hint: break down A into columns and B into rows, and use sub-problem 6.)
8. 🐛 Prove that for any square matrices A, B , then $AB - BA$ CANNOT be the identity matrix. (However, in infinite dimensional spaces, $AB - BA = I$ is related to the famous Heisenberg uncertainty principle in physics.) (Hint: Obviously use previous subproblems.)

Problem 6.2. A permutation is a way to rearrange the orders of objects. For example, if we have three objects 1, 2, 3, then a permutation is a bijective map $\sigma : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$. Maybe it sends 1, 2, 3 to 2, 1, 3 respectively. This concept is very important in games and computer science.

One way to study a permutation is to study a corresponding matrix. A permutation matrix is a linear map that permutes the standard basis vectors. For example, $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ corresponds to the permutation $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) \mapsto (\mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_3)$. We see that $\mathbf{e}_3$ is unchanged, so this is a **fixed point** of our permutation. In this example, we have one fixed point in the standard basis vectors.

1. Find a 5×5 permutation matrix P such that $P^k \neq I$ for all $1 \leq k \leq 5$, but $P^6 = I$.
2. For a permutation matrix P , prove that $\text{tr}(P)$ is the number of fixed standard basis vectors for the corresponding permutation.
3. 🐛 Consider the compositions of two permutations $P_1 P_2$ and $P_2 P_1$. Show that they always have the same number of fixed standard basis vectors. (Hint: Look at the trace problem above.)

Problem 6.3. Which of the following are linearly independent? If dependent, point out a non-zero linear relation. If independent, prove it.

1. (No credit, just format.) $1, x, x^2$. This is linearly independent. Suppose we have a linear relation

$$(1, x, x^2) \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0, \text{ the zero function. Then plug in } x = 0, 1, 2, \text{ we get three equations}$$

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0,$$

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0,$$

$$\begin{bmatrix} 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0.$$

This means

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \mathbf{0}.$$

Solve the equation by Gaussian elimination, and you will see that the only solution is $\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \mathbf{0}$. So the only linear relation is the zero relation, i.e., they are linearly independent.

2. $\sin x, \sin(x+1), \sin(x+2)$.
3. $\sin x, \sin(2x), \cos(x), \cos(2x)$.
4. e^x, e^{x+1}, e^{x+2} .
5. e^x, e^{2x}, e^{3x} .
6. $x^2, (x+1)^2, (x+2)^2$.
7. $(\sin x)^2, \cos(2x), 1$.
8. $1+i, 1-i$. (The field of scalars is $\mathbb{R}$.)
9. $1+i, 1-i$. (The field of scalars is $\mathbb{C}$.)
10. $1, \pi$. (The field of scalars is $\mathbb{R}$.)
11. $1, \pi$. (The field of scalars is $\mathbb{Q}$.)
12. $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$.
13. Burger, Wing, Coke, and a combo (made of one burger and two wings).
14. Burger, Wing, Coke, and Nothing. (Here “nothing” is the zero vector.)

Problem 6.4 (Vector spaces and basis). (This problem requires you to know the axioms of a vector space in the optional section.)

For the following, if it is not a vector space, just point out which axiom of vector spaces is failed by what vectors. If it is a vector space, you don’t need to verify axioms. Just find its dimension instead, and find a basis if it is finite dimensional. (All spaces are over $\mathbb{R}$ unless specifically mentioned.)

1. V is the set of all discontinuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ with its usual addition and scalar multiplication.
2. The solution set of $x^2 + y^2 \leq z^2$ in $\mathbb{R}^3$. (This is the pair of cones but we also include the insides of the two cones.)

3. ♣(Boolean algebra for computer science) The set of subsets of $X = \{1, 2, 3\}$, where we define $S + T$ for subsets S, T as the *symmetric difference*, i.e., $S + T = \{x \in X \mid x \in S \text{ but not } T, \text{ or } x \in T \text{ but not } S\}$.

This is a vector space defined over $\mathbb{F}_2$, where the vectors are subsets, and scalar-vector multiplication is simply $0S = \emptyset$ and $1S = S$ for all subsets S .

(Here $\mathbb{F}_2 = \{0, 1\}$ such that $0 = 0 + 0 = 1 + 1 = 0 \times 1 = 1 \times 0 = 0 \times 0$ and $1 = 0 + 1 = 1 + 0 = 1 \times 1$. You can also think “even” and “odd” in place of 0 and 1. Then all these calculations would make sense.)

4. $J = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and V is the space of all matrices that commutes with J . (You can use the results of previous HW directly.)

5. $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and V is the space of all matrices that commutes with A .

6. ♣ V is the space of all 3×3 magic matrices. (I.e., matrices like the zero matrix, or like $M = \begin{bmatrix} 2 & 9 & 4 \\ 7 & 5 & 3 \\ 6 & 1 & 8 \end{bmatrix}$,

where each row, each column, and each of the two diagonal adds up to the same number.)

(Hint: Flip all columns of M . Flip all rows of M . Take transpose of M . Take reflection of M along the anti-diagonal. Take differences of these things. Take linear combinations of these things. Try until you figure this out.)

Problem 6.5 (Solution space to a differential equation). Consider the space V of all real functions f that solves the differential equation $f''' - 6f'' + 11f' - 6f = 0$. You may use the fact that V is spanned by e^x, e^{2x}, e^{3x} .

1. Show that the matrix $A = \begin{bmatrix} e^0 & e^0 & e^0 \\ e^1 & e^2 & e^3 \\ e^2 & e^4 & e^6 \end{bmatrix}$ is invertible.
2. Show that e^x, e^{2x}, e^{3x} are linearly independent, so that they form a basis for V . (Hint: if they are dependent, then what does that tell you about the matrix A in the last subproblem?)
3. ♣Given any $f \in V$, find the coordinates of f under this basis in terms of $f(0), f(1), f(2)$. (Hint: use the matrix A in the first sub-problem. You may simply write A^{-1} without calculating the inverse matrix.)

Problem 6.6 (Equation space and its subspace). let V be the space of linear equations on three variables, with elements such as $ax + by + cz = d$ for arbitrary real numbers a, b, c, d . So the vectors here are actually equations. We can do linear combination of these equations, i.e., $2(2x + 3y - z = 3) + 3(x - y = 2) = (7x + 3y - 2z = 12)$.

(Note that since $x = 1$ and $x = 2$ are both elements of V , and V is a vector space, therefore V must include its difference, i.e., $0 = 1$. This equation $0 = 1$ is also an element of V , and it is indeed of the form $ax + by + cz = d$ where $a = b = c = 0$ and $d = 1$.)

1. Find a basis of V , and find the dimension of V .
2. Suppose we have a basis made of equations $\mathbf{v}_1, \dots, \mathbf{v}_k$. What is the solution set when all equations $\mathbf{v}_1, \dots, \mathbf{v}_k$ are satisfied?
3. Fix a point $\mathbf{p}$ in $\mathbb{R}^3$. Let W be the subset of V , made of all equations that contains this point $\mathbf{p}$ in its solution set. Is W a subspace? Prove or give counter examples. (Hint: to show that something is a subspace, you just need to show that $\mathbf{v} + \mathbf{w}$ and $k\mathbf{v}$ are still in W when $\mathbf{v}, \mathbf{w}$ is in W .)

4. Consider $x = 1, y = 2, z = 3, x + y + z = 7$. Are they linearly independent? Prove or provide a linear relation. (Note: you might want to try this proof for fun. First prove that the first three are linearly independent. For the last equation, take the subspace of all equations whose solution set contains the point $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Then this subspace contains $x = 1, y = 2, z = 3$, but fails to contain the last equation $x + y + z = 7$. This gives you a way to use independence extension lemma.)
5. Consider $x = 1, x = 2, x = 3$. Are they linearly independent? Prove or provide a linear relation.
6. (Not part of the HW) If you are curious, you may consider ways to generalize the statements here. What if we fix a line L , and let W be made of all equations that contains this line L in its solution set? What if we include more variables and go to higher dimensions?

7 HW Due Apr 28th (Later than usual. Use these to review for midterm)

Problem 7.1. There is a fast way to find a basis for the kernel. Please read example 5.7.1 in our lecture notes “EZ”. No need to read the proof, you can just use the method described directly.

For this problem, find a basis for the following spaces.

1. $\text{Ker}\left(\begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}\right).$
2. $\text{Ker}\left(\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}\right).$
3. $\text{Ker}\left(\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}\right).$
4. $\text{Ker}\left(\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & 1 & -3 \\ 0 & 1 & 2 & 2 & 6 \\ 5 & 4 & 3 & 3 & 1 \end{bmatrix}\right).$
5. $\text{Ker}\left(\begin{bmatrix} 1 & -2 & 1 & 1 & -1 \\ 2 & 1 & -1 & -1 & -1 \\ 1 & 7 & -5 & 5 & 5 \\ 3 & -1 & -2 & 1 & -1 \end{bmatrix}\right).$
6. $\text{Ker}\left(\begin{bmatrix} 2 & 1 & -1 & -1 & -1 \\ 1 & -1 & 1 & 1 & -2 \\ 3 & 3 & -3 & -3 & 4 \\ 4 & 5 & -5 & -5 & 7 \end{bmatrix}\right).$
7. $\text{Ker}\left(\begin{bmatrix} I_{n \times n} & I_{n \times n} \end{bmatrix}\right).$
8. $\text{Ker}\left(\begin{bmatrix} I_{n \times n} & I_{n \times n} & I_{n \times n} \end{bmatrix}\right).$
9. $\text{Ker}\left(\begin{bmatrix} I_{n \times n} & I_{n \times n} \\ O & O \end{bmatrix}\right).$

Problem 7.2. Find the matrices for the following linear maps under the given basis.

1. V is the span of functions $1, \frac{1}{x}, x \sin(\frac{1}{x})$, and the linear map is $\lim_{x \rightarrow \infty} : V \rightarrow \mathbb{R}$.
2. V is the space of 2×2 skew symmetric matrices, i.e., matrices such that $A^T = -A$, and T is the transpose map from V to V . Pick any basis for V .
3. V is the span of functions $1, x + 1, x^2$, and $D : V \rightarrow V$ is the derivative map.
4. V is the space of 2×3 matrices with basis $e_i e_j^T$. The linear map sends X to $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} X$.

Problem 7.3. For the following vector spaces V and their basis $\mathcal{B}, \mathcal{C}$, please find the change of coordinate matrix from $\mathcal{B}$ to $\mathcal{C}$.

1. V is the span of $\mathcal{B} = (1, x, x^2, x^3)$. $\mathcal{C} = (1, x + 2, (x + 2)^2, (x + 2)^3)$.

2. V is the span of $\mathcal{B} = (\sin(x), \sin(x+1))$. $\mathcal{C} = (\sin(x+2), \sin(x+3))$. (Hint: maybe use $(\sin x, \cos x)$ as an intermediate basis between the transition.)

3. V is the space of 2×3 matrices. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$.

$$\mathcal{B} = (A, [I_{2 \times 2} \quad \mathbf{0}], [\mathbf{0} \quad I_{2 \times 2}], BA, AC, AC^2).$$

$$\mathcal{C} = (e_1 e_1^T, e_1 e_2^T, e_1 e_3^T, e_2 e_1^T, e_2 e_2^T, e_2 e_3^T).$$

4. V is the space of 2×2 symmetric matrices.

$$\mathcal{B} = \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right).$$

$$\mathcal{C} = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \right).$$

5. $V = \mathbb{R}^3$. Vectors in $\mathcal{B}$ are columns of an invertible matrix B . Vectors in $\mathcal{C}$ are columns of an invertible matrix C . Express your answer in terms of B and C .

Problem 7.4. Find the basis for $\text{Ran}(A)$, $\text{Ran}(A^T)$, $\text{Ker}(A)$, $\text{Ker}(A^T)$, where we have

$$A = \begin{bmatrix} 1 & & & \\ 6 & 1 & & \\ 9 & 8 & 1 & \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ & 1 & 2 & 3 \\ & & 1 & 2 \end{bmatrix}.$$

Problem 7.5. Suppose the solution set to $A\mathbf{x} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ is $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} + k \begin{bmatrix} 0 \\ 1 \end{bmatrix} : k \in \mathbb{R} \right\}$. Find A .

Problem 7.6 (Space of matrices). Consider $M_{2 \times 2}$ be the space of 2×2 matrices. We fix a basis $X_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $X_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $X_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $X_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. We also fix a matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

- Consider the linear map $L_A : M_{2 \times 2} \rightarrow M_{2 \times 2}$ that sends X to AX . Write out the matrix for this linear map under the given basis. (Preferably as a block matrix. The block form would make it easier to imagine generalizations to higher dimensions.)
- Consider the linear map $R_A : M_{2 \times 2} \rightarrow M_{2 \times 2}$ that sends X to XA . Write out the matrix for this linear map under the given basis. (Preferably as a block matrix. The block form would make it easier to imagine generalizations to higher dimensions.)
- Calculate the 4×4 matrix multiplication $L_A R_A$ and $R_A L_A$. (You may use block matrix multiplication to facilitate the process.)
- Can you see why $L_A R_A = R_A L_A$ without any calculation? (Hint: Think about the meaning of these linear maps. L_A means multiplying A to X from the left, while R_A means multiplying A to X from the right. Which law of matrix multiplication is this?)
- Find a basis of $M_{2 \times 2}$ made of invertible matrices, and find the change of coordinate matrix to this new basis.

Problem 7.7 (Change of basis and linear maps). Let V be a space with three bases $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ and W be a space with two bases $\mathcal{C}_1, \mathcal{C}_2$, and a bijective linear map $L : V \rightarrow W$. We use $I_{j \leftarrow i}$ to mean a change of coordinate map from $\mathcal{B}_i$ to $\mathcal{B}_j$ or from $\mathcal{C}_i$ to $\mathcal{C}_j$, and $L_{j \leftarrow i}$ to mean the matrix for L from $\mathcal{B}_i$ to $\mathcal{C}_j$. Which of the following formula about matrices $L_{j \leftarrow i}, I_{j \leftarrow i}$ are true? (No proof needed for correct ones. But for wrong ones, just briefly point out what is wrong with it.)

1. $L_{1 \leftarrow 1} = I_{1 \leftarrow 2} L_{2 \leftarrow 1}$.
2. $L_{1 \leftarrow 1}^{-1} = L_{2 \leftarrow 1}^{-1} I_{1 \leftarrow 2}$.
3. $L_{2 \leftarrow 2} = I_{2 \leftarrow 1} L_{1 \leftarrow 1} I_{2 \leftarrow 1}^{-1}$.
4. $L_{2 \leftarrow 2} = L_{2 \leftarrow 1} I_{1 \leftarrow 1} L_{1 \leftarrow 2}$.
5. $L_{1 \leftarrow 3} = I_{1 \leftarrow 2} L_{2 \leftarrow 1} I_{1 \leftarrow 3} I_{3 \leftarrow 2} L_{1 \leftarrow 2}^{-1} I_{1 \leftarrow 2} L_{2 \leftarrow 1} I_{1 \leftarrow 2} I_{2 \leftarrow 3}$.

Problem 7.8. Given arbitrary subspaces of the domain and the codomain, could they be the four fundamental subspaces of some matrix?

Let V be any subspace of $\mathbb{R}^n$, and W be any subspace of $\mathbb{R}^m$, and $\dim V = \dim W = d$. We want to find a matrix whose four fundamental subspaces are $V, V^\perp, W, W^\perp$.

1. Let B be an $n \times d$ matrix whose column vectors form a basis for V . Show that $\text{Ran}(B) = V$ and $\text{Ker}(B^T) = V^\perp$, and $\text{Ker}(B) = 0$.
2. Let C be an $m \times d$ matrix whose column vectors form a basis for W . Show that $\text{Ran}(C) = W$ and $\text{Ker}(C^T) = W^\perp$, and $\text{Ker}(C) = 0$.
3. Show that $CB^T \mathbf{x} = \mathbf{0}$ if and only if $B^T \mathbf{x} = \mathbf{0}$. Similarly, $BC^T \mathbf{x} = \mathbf{0}$ if and only if $C^T \mathbf{x} = \mathbf{0}$.
4. Show that CB^T is the desired matrix, and in fact CB^T gives a full rank decomposition.

Problem 7.9 (Fredholm's Alternative). This is an interesting result with some future application in partial differential equations.

We have two problems. Problem One is to solve the equations $A\mathbf{x} = \mathbf{b}$. Problem Two is to solve the equations $\begin{bmatrix} A^T \\ \mathbf{b}^T \end{bmatrix} \mathbf{y} = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix}$.

Fredholm's Alternative is a theorem that states that: exactly one of these two problems has a solution. Not both, not neither.

1. Suppose $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 3 & 4 & 5 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 5 \\ 5 \\ 9 \end{bmatrix}$. Verify Fredholm's Alternative here: $A\mathbf{x} = \mathbf{b}$ has no solution, while $\begin{bmatrix} A^T \\ \mathbf{b}^T \end{bmatrix} \mathbf{y} = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix}$ has a solution.
2. In the subproblem above, if $\mathbf{y}$ is a solution, then show that the equations in $A\mathbf{x} = \mathbf{b}$ can be linearly combined into the equation $0 = 1$, by using coordinates of $\mathbf{y}$ as a coefficient. (This serves as an intuition for Fredholm's Alternative.)
3. If $\mathbf{b} \in \text{Ran}(A)$, show that Problem One has a solution, while Problem Two has no solution.
4. If $\mathbf{b} \notin \text{Ran}(A)$, show that Problem One has no solution, while Problem Two has a solution. (Hint: consider the last non-zero row of $\text{RREF}([A \quad \mathbf{b}])$.)

Problem 7.10. Given a subspace V of $\mathbb{R}^n$, how to find its orthogonal complement? First, we find a basis $A = [\mathbf{v}_1 \ \dots \ \mathbf{v}_k]$ for V . Then note that $V^\perp = \text{Ran}(A)^\perp = \text{Ker}(A^T)$. Therefore, we can use the kernel finding algorithm/ Gaussian elimination to find a basis for $\text{Ker}(A^T)$.

1. Find the orthogonal complement of the span of $\begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$

2. In $\mathbb{R}^3$, find a basis for the orthogonal complement of $\text{span}\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right)$ where $a \neq 0$. (This is a basis of the

subspace with normal vector $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$.)

3. The chess matrix is

$$A = \begin{bmatrix} r & n & b & q & k & b & n & r \\ p & p & p & p & p & p & p & p \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p & p & p & p & p & p & p & p \\ r & n & b & q & k & b & n & r \end{bmatrix}.$$

Here r, n, b, q, k, p are distinct non-zero real numbers. Find a basis for all four fundamental subspaces of A .

4. Suppose $A = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} I & G \\ 0 & 0 \end{bmatrix}$ where F, G are both $n \times n$, and the four fundamental subspaces of A and B are identical. Prove that $F = G$.

Problem 7.11 (More Inclusion-Exclusion Principals). (No credit, just for fun)

1. Prove that for subsets X, Y, Z , we have $|X \cup Y \cup Z| = |X| + |Y| + |Z| - |X \cap Y| - |X \cap Z| - |Y \cap Z| + |X \cap Y \cap Z|$.
2. Find an example of subspaces X, Y, Z such that $\dim(X + Y + Z) \neq \dim(X) + \dim(Y) + \dim(Z) - \dim(X \cap Y) - \dim(X \cap Z) - \dim(Y \cap Z) + \dim(X \cap Y \cap Z)$.
3. Prove that for subspaces X, Y, Z , we have $\dim(X + Y + Z) \leq \dim(X) + \dim(Y) + \dim(Z) - \dim(X \cap Y) - \dim(X \cap Z) - \dim(Y \cap Z) + \dim(X \cap Y \cap Z)$. (Hint: Use the regular IEP for subspaces repeatedly. Prove that $(X \cap Z) + (Y \cap Z) \subseteq (X + Y) \cap Z$.)

Problem 7.12 (Meaning of rank-one decomposition). (No credit, just for fun)

Suppose we have seven people, A, B, ..., G, and they watched six videos on bilibili, a, b, ..., f. Suppose for each video, they can click "like" or "dislike", and this data is collected into the following matrix.

$$M = \begin{bmatrix} & a & b & c & d & e & f \\ A & 1 & 1 & 1 & 0 & 0 & 0 \\ B & 1 & 1 & 1 & 0 & 0 & 0 \\ C & 1 & 1 & 1 & 0 & 0 & 0 \\ D & 1 & 1 & 1 & 1 & 1 & 1 \\ E & -1 & -1 & -1 & 1 & 1 & 1 \\ F & -1 & -1 & 1 & 1 & 1 & 1 \\ G & -1 & -1 & -1 & 1 & 1 & 1 \end{bmatrix}.$$

Here 1 means the corresponding person likes the corresponding video, -1 means dislike, and 0 means no response.

Let us suppose that a, b, c are all educational videos, while c, d, e, f are all funny videos. So we have two categories for a video: educational or funny. Note that each person seems to react reasonably consistent about videos in each category. The exception is c , which is both educational and funny, and whether a person likes or dislikes seems to be influenced by other factors.

1. What is the rank of this 7×6 matrix?

2. Build a matrix X such that its two columns corresponds to the two categories of videos, and its seven rows corresponds to the seven people, and the entries records whether the corresponding person likes, dislikes or has no response to the corresponding category in general.
3. Build a matrix Y such that its two rows corresponds to the two categories of videos, and its six columns corresponds to the six videos. And an entry is 1 if the video is in this category, zero if this video is NOT in this category.
4. Check to see that $M - XY$ has rank 1. (This represents portions of M that cannot be explained by categories of the videos.)
5. 🍷 Write M as the sum of three rank one matrices, where the first one reflects how educational videos contributes to M , the second one reflects how funny videos contributes to M , and the third one reflects how other factors contributes to M .

8 HW Due May 12th (Later than usual. Happy Holiday)

Problem 8.1 (Orthogonal basis made of coordinates ± 1). A **Hadamard matrix** is a matrix H whose entries are all ± 1 , and the columns are mutually orthogonal (but they don't have to be unit vectors). (When n is a power of 2, then an example of Hadamard matrix would be the matrix made by the Haar wavelet basis.)

1. For any $n \times n$ Hadamard matrix, compute $H^T H$.
2. If H is a Hadamard matrix, show that $\begin{bmatrix} H & H \\ H & -H \end{bmatrix}$ is also a Hadamard matrix.
3. Can you find a 3×3 Hadamard matrix? Find it or show why not.
4. Let H be any Hadamard matrix. Then if we permute the rows and columns, or if we negate some rows and columns (i.e., multiply the row or column by -1), prove that the result is still a Hadamard matrix. (If two Hadamard matrices can be obtained from each other like this, then we say they are equivalent.)
5. (No Credit, just for fun) Prove that all 4×4 Hadamard matrices are equivalent.
6. (Read only) It is conjectured that a Hadamard matrix should exist for all $(4k) \times (4k)$ matrices. However, this remains unproven to this date. For example, is there a 668×668 Hadamard matrix? We don't know the answer yet, as far as I can tell. The Haar wavelet basis gives a Hadamard matrix whenever n is a power of 2. But, for example, try to find a Hadamard matrix when $n = 12$, if you want a challenge. And on the question of equivalence, in general, Hadamard matrices of the same size could be inequivalent. For example, there are five inequivalent 16×16 Hadamard matrices.

Problem 8.2 (A useful “non-inner product” space). Consider the space $\mathbb{R}^4$ where we define $\left\langle \begin{bmatrix} x_1 \\ y_1 \\ z_1 \\ t_1 \end{bmatrix}, \begin{bmatrix} x_2 \\ y_2 \\ z_2 \\ t_2 \end{bmatrix} \right\rangle = x_1 x_2 + y_1 y_2 + z_1 z_2 - t_1 t_2$. This is NOT an inner product, but nevertheless, this is the structure used by special relativity. This is sometimes called a **Minkowski space-time**. We say $\mathbf{v} \perp \mathbf{w}$ if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$ under the definition above.

1. Find a matrix D such that $\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}^T D \mathbf{w}$. Is D symmetric? Positive Definite?
2. For an arbitrary subspace $W \subseteq \mathbb{R}^4$, we define its “orthogonal complement” to be $W^\perp = \{\mathbf{v} \in \mathbb{R}^4 \mid \mathbf{v} \perp W\}$. Show that we always have $\dim W + \dim W^\perp = 4$. (Hint: use rank-nullity.)
3. Find a subspace strictly contained in its own “orthogonal complement”. (“strictly contained” means they are not equal.)
4. Find a basis for $\mathbb{R}^4$ made of vectors of “length” zero. (Length zero means $\sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = 0$ in this Minkowski space-time.)

Problem 8.3. Consider the following functions.

1. $f(x, y, z) = 2x^2 + 4y^2 + 3z^2 + 4xy + 4yz$. Find a symmetric matrix A such that $f(x, y, z) = \begin{bmatrix} x & y & z \end{bmatrix} A \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.
Is A positive definite? If so, find the Chelosky decomposition of A , and then complete the squares for $f(x, y, z)$.

2. $f(x, y) = 4x^2 + 4xy + 2y^2 + 2y + 1$. Find a symmetric matrix A such that $f(x, y) = \begin{bmatrix} x & y & 1 \end{bmatrix} A \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$.

Is A positive definite? If so, find the Chelosky decomposition of A , and then complete the squares for $f(x, y)$.

Problem 8.4. Suppose $\mathbf{a}_1, \dots, \mathbf{a}_n$ form an orthonormal basis for an inner product space V . Either prove that the followings are orthonormal basis, or perform Gram-Schmidt to them.

1. $n = 3$ and we have $\mathbf{b}_1 = \frac{1}{3}(2\mathbf{a}_1 + 2\mathbf{a}_2 - \mathbf{a}_3)$, $\mathbf{b}_2 = \frac{1}{3}(2\mathbf{a}_1 - \mathbf{a}_2 + 2\mathbf{a}_3)$, $\mathbf{b}_3 = \frac{1}{3}(-\mathbf{a}_1 + 2\mathbf{a}_2 + 2\mathbf{a}_3)$.
2. $n = 5$, and $\mathbf{b}_1 = \mathbf{a}_1 + \mathbf{a}_5$, $\mathbf{b}_2 = \mathbf{a}_1 - \mathbf{a}_2 + \mathbf{a}_4$, $\mathbf{b}_3 = 2\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3$.

Problem 8.5. Find bases to the following subspaces V and their orthogonal complement $V^\perp$ of $\mathbb{R}^4$. And write $\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2$ where $\mathbf{b}_1 \in V$, $\mathbf{b}_2 \in V^\perp$.

1. V is the hyper plane by the equation $x_1 + x_2 + x_3 - x_4 = 0$. And the vector is $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$.

2. V is the span of $\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2$. And the vector is $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \end{bmatrix}$.

3. V is the span of $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$. And the vector is $\mathbf{b} = \begin{bmatrix} 4 \\ 0 \\ -1 \\ 1 \end{bmatrix}$.

Problem 8.6. Can we do QR to matrices where the columns are dependent? Let us see a special case.

Say $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where a, c are not both zero. Suppose we have a QR decomposition $A = QR$

1. Prove that

$$Q = \frac{1}{\sqrt{a^2 + c^2}} \begin{bmatrix} a & -c \\ c & a \end{bmatrix}, R = \frac{1}{\sqrt{a^2 + c^2}} \begin{bmatrix} a^2 + c^2 & ab + cd \\ 0 & ad - bc \end{bmatrix}.$$

2. Find the QR decomposition of $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ using the formula above. For the latter one, note that R is not invertible.
3. If A is 3×3 , this formula is much harder to do, and too complicated to be useful. But at least the formula for the first column of Q should be easy, right?

Problem 8.7. Let $Q = k \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}$.

1. Find k such that Q is an orthogonal matrix.
2. Find Q^{-1} . (Here k is unknown and Q might not be an orthogonal matrix.)

3. Let $W = \frac{1}{2} \begin{bmatrix} 1 & 1 & \sqrt{2} & 0 \\ 1 & 1 & -\sqrt{2} & 0 \\ 1 & -1 & 0 & \sqrt{2} \\ 1 & -1 & 0 & -\sqrt{2} \end{bmatrix}$. And let's say k is chosen such that Q is an orthogonal matrix.

Then columns of Q and columns of W are two bases of $\mathbb{R}^4$. Find $I_{W \leftarrow Q}$ and $I_{Q \leftarrow W}$.

Problem 8.8. For the following matrices, find their QR decomposition.

1. $\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix}$.

2. $\begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$.

3. $\begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix}$.

4. $\begin{bmatrix} 1 & 1 & -1 \\ 2 & -1 & 2 \\ -1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.

9 HW Due May 19

Problem 9.1. Let $Q = k \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}$.

1. Find the orthogonal projection of $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ onto the span of the first column of Q .
2. Find the orthogonal projection of $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ onto the span of the first two columns of Q .

Problem 9.2. Find the following projection matrices

1. Find the orthogonal projection to the plane $x - y - 2z = 0$ in $\mathbb{R}^3$.
2. Find the orthogonal projection to the orthogonal complement of the plane $x - y - 2z = 0$ in $\mathbb{R}^3$.
3. Find the orthogonal projection to the range of $\begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix}$.
4. Find the orthogonal projection to the kernel of $\begin{bmatrix} 3 & 6 & 6 \\ 4 & 8 & 8 \end{bmatrix}$. (Hint: maybe first find the orthogonal projection to $\text{Ran}(A^T)$.)
5. Find the orthogonal projection to the range of $\frac{1}{6} \begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}$.

Problem 9.3. Calculate the following determinants.

1. $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & -1 & 2 & 1 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$.

2. $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & -1 & 1 & -1 \end{bmatrix}$. (This is a Vandermonde matrix for $1, 2, 3, -1$. We will talk about this in later classes. The answer should be $\pm(3-2)(3-1)(3-(-1))(2-1)(2-(-1))(1-(-1))$, i.e., plus or minus the product of all possible differences of these values.)

3. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{bmatrix}$. (This is an anti-circulant matrix.)

4. $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 11 & 12 & 0 & 0 & 0 \\ 13 & 14 & 0 & 0 & 0 \\ 15 & 16 & 0 & 0 & 0 \end{bmatrix}$. (Hint: Don't calculate. Just stare at it until you see the answer.)

5.
$$\begin{bmatrix} 0 & \dots & 0 & 1 \\ \vdots & \ddots & \ddots & 0 \\ 0 & \ddots & \ddots & \vdots \\ 1 & 0 & \dots & 0 \end{bmatrix}.$$
 The matrix is $n \times n$.

Problem 9.4. Let A be a 3×3 matrix $\begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \mathbf{a}_3^T \end{bmatrix}$ with $\det(A) = 5$. Find the determinant of the matrices below.

1. $2A$.
2. $-A$.
3. A^2 .
4. A^{-1} .
5. A^T .
6. $\begin{bmatrix} \mathbf{a}_1^T - \mathbf{a}_3^T \\ \mathbf{a}_2^T - \mathbf{a}_1^T \\ \mathbf{a}_3^T - \mathbf{a}_2^T \end{bmatrix}$. (Hint: Do this in two ways if you like. You can do this with row operations, or via decomposition $\det(EA) = \det(E)\det(A)$.)
7. $\begin{bmatrix} \mathbf{a}_1^T + \mathbf{a}_3^T \\ \mathbf{a}_2^T + \mathbf{a}_1^T \\ \mathbf{a}_3^T + \mathbf{a}_2^T \end{bmatrix}$. (Hint: Do this in two ways if you like. You can do this with row operations, or via decomposition $\det(EA) = \det(E)\det(A)$.)

Problem 9.5. Let $A = [\mathbf{a}_1 \ \dots \ \mathbf{a}_n]$ be any $n \times n$ matrix. Geometrically speaking, $|\det(A)|$ is the absolute volume of the parallelotope A , and $\|\mathbf{a}_i\|$ is the length of the i -th edge. As you can imagine, the volume $|\det(A)|$ should be the product $\|\mathbf{a}_1\| \dots \|\mathbf{a}_n\|$ when all edges are mutually orthogonal. And when the edges are NOT mutually orthogonal, then intuitively $|\det(A)|$ should be strictly less than the product $\|\mathbf{a}_1\| \dots \|\mathbf{a}_n\|$.

This is the famous Hadamard inequality $|\det(A)| \leq \|\mathbf{a}_1\| \dots \|\mathbf{a}_n\|$. We prove this inequality in this problem.

1. Suppose A is invertible and we have QR decomposition $A = QR$ where $R = [\mathbf{r}_1 \ \dots \ \mathbf{r}_n]$ is upper triangular. Show that $\|\mathbf{r}_i\| = \|\mathbf{a}_i\|$. (Hint: Q is an orthogonal matrix, so it preserves length.)
2. Show that in the set up above, $\det(R) \leq \|\mathbf{r}_1\| \dots \|\mathbf{r}_n\|$.
3. Prove the Hadamard inequality $|\det(A)| \leq \|\mathbf{a}_1\| \dots \|\mathbf{a}_n\|$.

10 HW Due May 26

Problem 10.1. Prove or find counter examples.

1. We always have $\det(AB - BA) = 0$ for any square matrices A, B .
2. We always have $\det(-A) = -\det(A)$ for square matrices A .
3. If n is odd and A is $n \times n$ and skew-symmetric, then A is not invertible. (Hint: take another look at the last sub-problem.)
4. Suppose we have LDU decomposition $A = LDU$. Let A_i be the upper left $i \times i$ block of A , and let d_i be the i -th diagonal entry of D . Then $d_i = \frac{\det(A_i)}{\det(A_{i-1})}$. (Hint: also write L, D, U in blocks.)

Problem 10.2 (Don't do these things). What's WRONG with these arguments?

1. For block matrix we have $\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(AD - BC)$. (Not part of the HW: What should the correct formula be if A is invertible?)
2. We have $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \det \left(\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \right) = \frac{ad-bc}{ad-bc} = 1$.
3. If $AB = -BA$, then $\det(A)\det(B) = -\det(B)\det(A)$, so $2\det(A)\det(B) = 0$. So one of A, B is not invertible.

Problem 10.3 (Leibniz formula). Use Leibniz formula (the big formula) to help with the following problems.

1. Consider $\det \begin{bmatrix} 2x & x & 1 & 2 \\ 1 & x & 1 & -1 \\ 3 & 2 & x & 1 \\ 1 & 1 & 1 & x \end{bmatrix}$. This is a polynomial in x . What is the coefficient for x^4 ? What is the coefficient for x^3 ?
2. Suppose we multiply each (i, j) entry of a 4×4 matrix A by j . For each term in the big formula of $\det(A)$, how would this term change? How would $\det(A)$ change?
3. Consider $A = \begin{bmatrix} a & 0 & b & 0 \\ 0 & c & 0 & d \\ e & 0 & f & 0 \\ 0 & g & 0 & h \end{bmatrix}$, where all letter variables are non-zero. In the big formula of $\det(A)$, how many non-zero terms are there? Can you factorize the determinant? (Hint: This is a "hidden" block diagonal matrix.)

Problem 10.4 (Pascal Matrices). Let P_n be the $n \times n$ symmetric Pascal's matrix. (E.g., the 4×4 version is

$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix}$.) Similarly, let L_n, U_n be the $n \times n$ lower triangular and upper triangular Pascal's matrix.

(E.g., the 4×4 version is $L_n = \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 2 & 1 & \\ 1 & 3 & 3 & 1 \end{bmatrix}$, and we always have $U_n = L_n^T$.)

For this problem, you may freely use the fact that P_n has LU decomposition $P_n = L_n U_n$.

1. Find $\det(P_n)$.

2. Let A_n be obtained by reducing the lower right entry of P_n by 1. Find $\det(A_n)$. (Hint: Can you see how cofactor is involved here?)

Problem 10.5 (Laplace expansion). Use Laplace expansion or cofactors to help with the following problems.

1. A Hessenberg matrix is a matrix that is “almost” triangular, except for an extra diagonal of en-

tries. For example, the matrices $H_4 = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$ is Hessenberg. Define the $n \times n$ matrix

$$H_n = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ & \ddots & \ddots & 1 \\ & & 1 & 2 \end{bmatrix}.$$

Show that $\det(H_n)$ is the Fibonacci sequence. (Hint: expand along the first column and establish the inductive property of the Fibonacci sequence, and verify the first two cases of $\det(H_n)$.)

2. A tridiagonal matrix is a matrix that is simultaneously upper Hessenberg and lower Hessenberg. For

example, $S_4 = \begin{bmatrix} 3 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 1 & 3 \end{bmatrix}$ is tridiagonal. Generalize S_4 into $n \times n$ matrices. Find the inductive

formula for $\det(S_n)$. How is this related to the Fibonacci sequence? (Hint: to see the relation to the Fibonacci sequence, maybe write out the first several terms and compare?)

11 HW Due June 2

Problem 11.1. Find all eigenvalues and all corresponding eigenvectors for the following matrices. (Including complex eigenvalues and eigenvectors.)

1. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$

2. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}.$

3. $\begin{bmatrix} 10 & 11 \\ 3 & 18 \end{bmatrix}.$

4. $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$

5. $\begin{bmatrix} b & -a \\ a & b \end{bmatrix},$ here $a, b \in \mathbb{R}.$

6. $\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 2 \\ -2 & -2 & 0 \end{bmatrix}.$

Problem 11.2. Find a formula for the sequence (a_n) under the given conditions. Try to describe the evolution using a matrix, and use the eigenvalues and eigenvectors of the matrix to compute the detail.

1. $a_0 = 0, a_1 = 1,$ and $a_{n+2} = 3a_{n+1} - 2a_n.$

2. $a_0 = 0,$ and $a_{n+1} = 2a_n + 1.$

3. $a_0 = 0,$ and $a_{n+1} = 2a_n + 2^n.$

4. $a_0 = 0, b_0 = 1, a_{n+1} = a_n + b_n, b_{n+1} = -2a_n + 4b_n.$

5. $a_0 = 0, a_1 = 1,$ and $a_{n+2} = 2a_{n+1} - a_n.$ (Hint: the matrix is similar to $\begin{bmatrix} 1 & k \\ 1 & 1 \end{bmatrix},$ whose power is also easy to compute.)

6. $a_0 = 0,$ and $a_{n+1} = a_n + n^2.$ (Hint: set $\mathbf{x}_n = \begin{bmatrix} a_n \\ n^2 \\ n \\ 1 \end{bmatrix}.$ The matrix is not diagonalizable though,

but if you write the matrix as $I + X,$ then what is $X^4?$ What does that tell you about a formula for $(I + X)^n?$)

Problem 11.3 (Eigenvector change). Let $p(x)$ be any polynomial. The intuition here is that any eigenvector of A must be an eigenvector of $p(A),$ but the converse is NOT true! In short, $p(A)$ sometimes could have more eigenvectors than $A.$ Let us verify some details. This problem categorized all possible reasons for this failure.

1. Suppose $A\mathbf{v} = \lambda\mathbf{v}.$ Show that $p(A)\mathbf{v} = p(\lambda)\mathbf{v}.$ (Hint: Prove for powers of A first. Then polynomials are linear combinations of powers.)

2. Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and $p(x) = (x-1)^2 + 1$. Find all eigenvalues of $p(A)$, and find a vector $\mathbf{v}$ which is an eigenvector of $p(A)$ but not an eigenvector of A . (Failure caused by killing non-diagonal portion through $p(x)$.)
3. Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and $p(x) = (x-1)(x-2) + 1$. Find all eigenvalues of $p(A)$, and find a vector $\mathbf{v}$ which is an eigenvector of $p(A)$ but not an eigenvector of A . (Failure caused by merging different eigenvalues into the same eigenvalue through $p(x)$.)

Problem 11.4 (Real Discrete Fourier Transform). Suppose A is a 4×4 matrix with diagonal entries 4 and non-diagonal entries -1 . Let H_4 be the matrix for the Haar wavelet basis.

1. Calculate AH_4 .
2. Calculate $H_4^{-1}AH_4$.
3. Find $\det(A)$, A^{-1} .

Problem 11.5 (Complex Discrete Fourier Transform). Consider $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$. Let $F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$.

(This F_4 is the 4×4 Fourier matrix, and it represents a discrete version of Fourier transform. Continuous Fourier transform is useful for the study of periodic functions, whereas here we shall see that discrete Fourier transform is useful for the study of “periodic matrices”.)

1. Calculate AF_4 .
2. Find the D, X in the diagonalization $A = XDX^{-1}$. (Hint: the last subproblem.)
3. Find $\det(A)$, A^{-1} .

Problem 11.6 (Pavel Grinfeld Problem). 🍷 Calculate $\begin{bmatrix} 110 & 55 & -164 \\ 42 & 21 & -62 \\ 88 & 44 & -131 \end{bmatrix}^{2017}$. (Hint: Stare at it, until you see two eigenvalues and their eigenvectors. If you cannot, stare harder! Then use trace to find the last eigenvalue.)

12 HW Due June 9

Problem 12.1. Find all possible values of the unknown constants given the condition.

1. $A = \begin{bmatrix} 0 & a \\ 1 & b \end{bmatrix}$ and the eigenvalues are 4, 7.
2. $A = \begin{bmatrix} 0 & 0 & a \\ 1 & 0 & b \\ 0 & 1 & c \end{bmatrix}$ and the eigenvalues are 2, 3, 4.
3. $A = \begin{bmatrix} 2 & -2 & 0 \\ -2 & x & -2 \\ -2 & -2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 2 & & \\ & 2 & \\ & & y \end{bmatrix}$, and A, B have the same characteristic polynomial.
4. $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & a \end{bmatrix}$ is invertible, and $\mathbf{x} = \begin{bmatrix} 1 \\ b \\ 1 \end{bmatrix}$ is an eigenvector of A .
5. $A = \begin{bmatrix} 3 & 2 & -2 \\ -k & -1 & k \\ 4 & 2 & -3 \end{bmatrix}$ is diagonalizable.

Problem 12.2 (Eigenvalue change). Suppose A has eigenvalues 1, 1, 2, 3. Find the eigenvalues of the following matrices.

1. $A - I$.
2. A^2 .
3. A^{-1} . (Why is A invertible?)
4. $\overline{A}$.
5. A^* .
6. $A^3 - 6A^2 + 11A - 6I$.
7. $\begin{bmatrix} A & \\ & A \end{bmatrix}$.
8. $\begin{bmatrix} A + I & A \\ & A - I \end{bmatrix}$.

Problem 12.3 (Large power is unstable). Large power of a large matrix is large? Prove the following problems. Hint: Diagonalization.

1. All entries of $\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}^{1024}$ are larger than 10^{700} . (You may use the fact that $\frac{1}{3}(5^{1024} - 2^{1024}) > 10^{700}$.)
2. $\begin{bmatrix} 3 & 2 \\ -5 & -3 \end{bmatrix}^{1024} = I_2$
3. $\begin{bmatrix} -5 & -7 \\ 3 & 4 \end{bmatrix}^{1024} = \begin{bmatrix} -5 & -7 \\ 3 & 4 \end{bmatrix}$.

4. (Read Only) You can use matrix calculator to verify that all entries of $\begin{bmatrix} -5 & -6.9 \\ 3 & 4 \end{bmatrix}^{1024}$ are less than 10^{-70} in absolute value. Furthermore, all entries of $\begin{bmatrix} -5 & -7.1 \\ 3 & 4 \end{bmatrix}^{1024}$ are larger than 10^{57} in absolute value. These things are extremely unstable....

Problem 12.4 (Eigenvalues are unstable). Tiny error in the entries could have drastic effect on eigenvalues and diagonalizabilities. Find the eigenvalues of the following matrices, and their algebraic and geometric multiplicity. Are they diagonalizable?

1. $\begin{bmatrix} 10 & 1 & \\ & 10 & 1 \\ & & 10 \end{bmatrix}$.
2. $\begin{bmatrix} 10 & 1 & \\ & 10.001 & 1 \\ & & 10.002 \end{bmatrix}$. (Note that diagonalizability could be different for the slightest change in entries.)
3. $\begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & 0 & 1 \\ & & & 0 \end{bmatrix}$.
4. $\begin{bmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & 0 & 1 \\ 0.0001 & & & 0 \end{bmatrix}$. (Note how the errors in entries is magnified in the errors in eigenvalues.)
5. (Read Only) In general, suppose you pick a random square matrix (all entries are random variables). Then it will have random eigenvalues $\lambda_1, \dots, \lambda_n$. When random variables are continuously distributed, the chances that two random variables are equal is zero. So a random matrix A (or anything with unknown tiny error) will have n distinct eigenvalues with probability 100%. In particular, a random matrix A is diagonalizable with probability 100%. In real life, if you obtain some square matrix with potential tiny errors, then you may simply assume that it is diagonalizable.

Problem 12.5. Find all eigenvalues, all algebraic multiplicity, all geometric multiplicity, and a description of all eigenvectors. Finally, find a diagonalization if possible, or find a Schur decomposition if it is not diagonalizable.

1. $A = \begin{bmatrix} a & b & \dots & b \\ b & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & b \\ b & \dots & b & a \end{bmatrix}$.
2. $A = \begin{bmatrix} 0 & -1 \\ 1 & -2 \end{bmatrix}$.

Problem 12.6. Are these diagonalizable?

1. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

$$2. A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 4 & 0 & 1 \end{bmatrix}.$$

3. $A = \mathbf{vw}^T$ for some $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$. Suppose $\mathbf{w}^T \mathbf{v} = 0$.

4. $A = \mathbf{vw}^T$ for some $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$. Suppose $\mathbf{w}^T \mathbf{v} \neq 0$.

Problem 12.7 (Polynomial identity and diagonalization). Let A be an $n \times n$ complex matrix. In general, if $p(A) = 0$ and $p(x)$ is a polynomial with distinct roots, then A is diagonalizable. Here we prove some special cases. (Note that we did similar things in class when $A^2 = I$ and when $A^2 = A$.)

1. Suppose $A^2 = 4I$. Show that $\text{Ker}(A + 2I) + \text{Ker}(A - 2I) = \mathbb{C}^n$, and A is diagonalizable. (Hint: If $(A + 2I)\mathbf{v} = \mathbf{0}$ and $(A - 2I)\mathbf{v} = \mathbf{0}$, take difference to see that we must have $\mathbf{v} = \mathbf{0}$.)
2. Suppose $A^2 - 3A + 2I = 0$. Show that $\text{Ker}(A - I) + \text{Ker}(A - 2I) = \mathbb{C}^n$, and A is diagonalizable.
3. Suppose $A^3 = A$. Find constant $a, b, c \in \mathbb{C}$ such that, for all $\mathbf{v} \in \mathbb{C}^n$, we have $\mathbf{v} = aA(A - I)\mathbf{v} + bA(A + I)\mathbf{v} + c(A - I)(A + I)\mathbf{v}$. (Maybe you've seen similar things in your calculus class called partial fractions?)
4. Suppose $A^3 = A$. Show that $\text{Ker}(A + I) + \text{Ker}(A - I) + \text{Ker}(A) = \mathbb{C}^n$, and A is diagonalizable.